MARKET CASE RECLINER APPLICATION

INTERNAL USE ONLY

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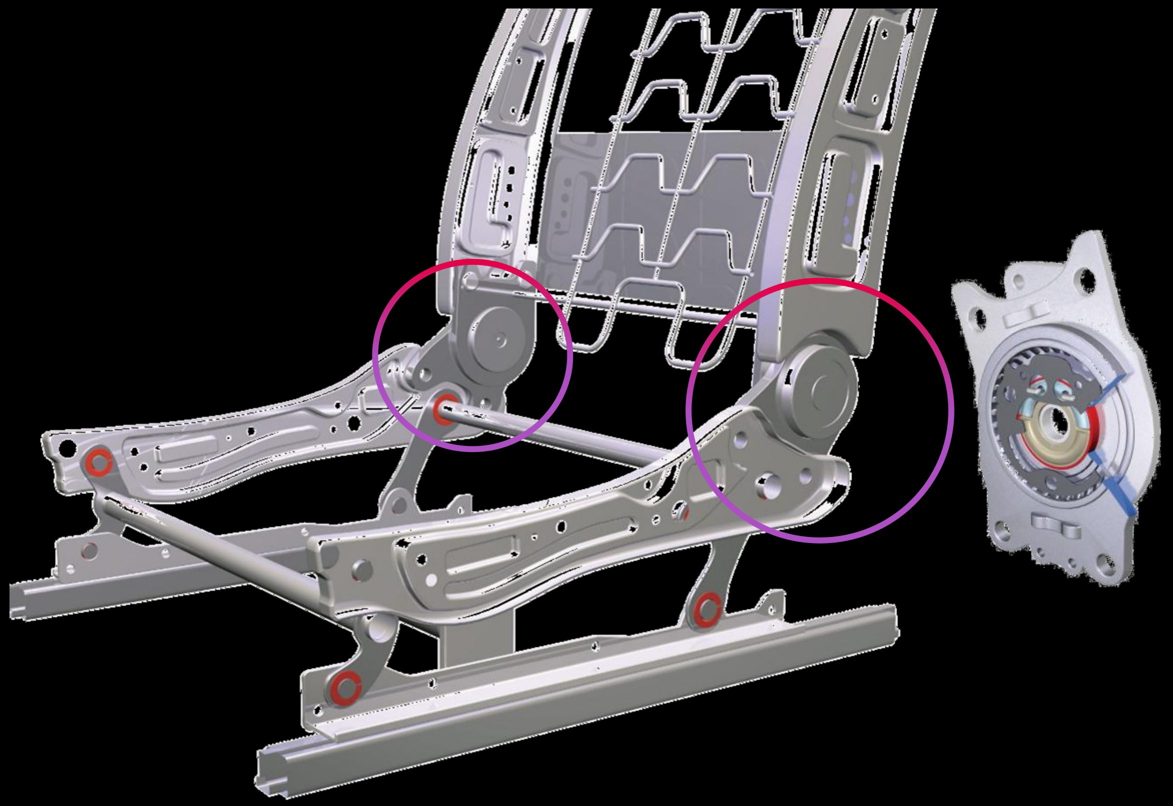
UNLOCKING YOUR ENGINEERING PUZZLES

A SAINT-GOBAIN BRAND

Description

What does a recliner

A recliner is the safety critical structural interface between the seat cushion frame and the backrest frame. It enables precise and controlled adjustment of the backrest angle while simultaneously withstanding extremely high loads during crash events.
Because the recliner forms the central mechanical link between the upper and lower seat structures, it plays a key role in load path management, occupant safety, and energy absorption in high severity impacts.



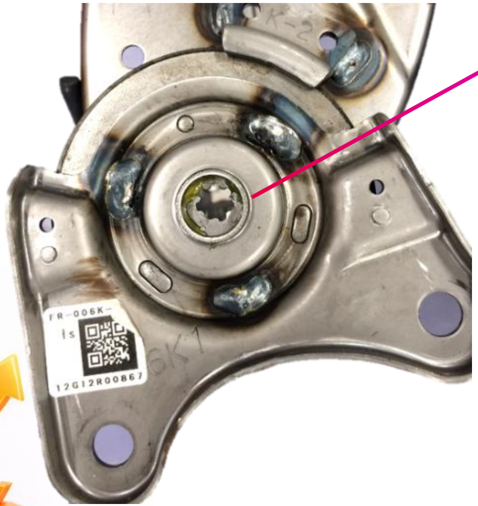
Source; Mazda Recliner Teardown

Technology Overview

- Recliner architecture

Manual Sector Gear Recliners
Core mechanism: Gear / sector gear combined with a pawl
Operation: Adjusted using a mechanical lever, with locking achieved through the shape of the teeth

Assembled condition



Stepless Knee Lever Recliners (Wedge Type)
Utilizes a knee lever mechanism for self-locking
Adjustment is achieved through force closure, supported by spring and wedge based mechanisms

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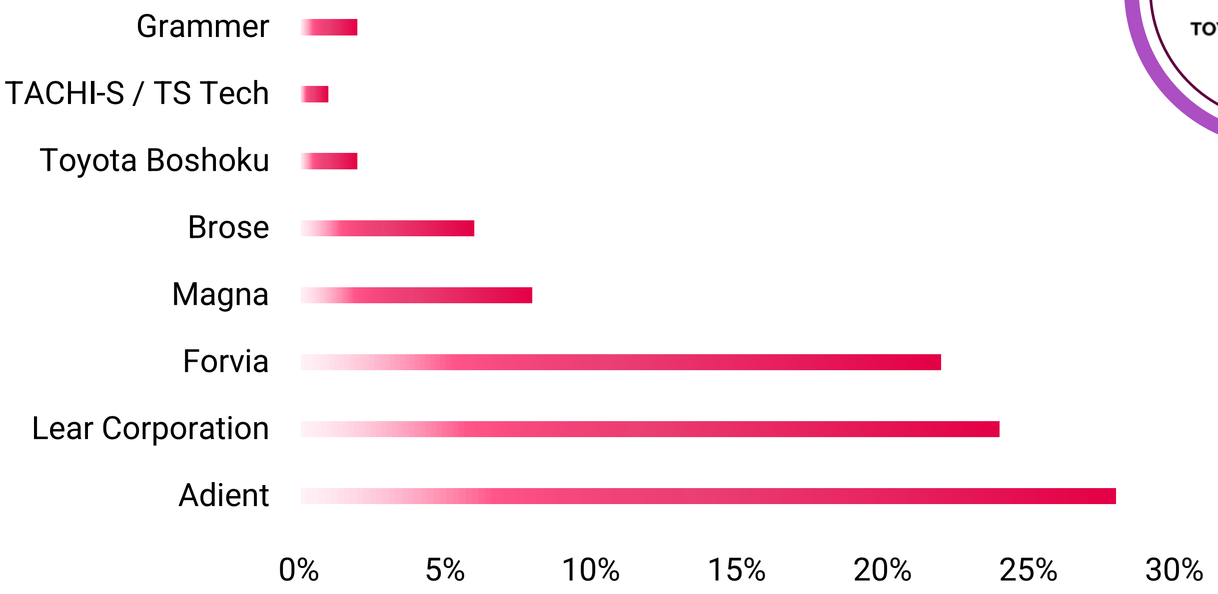
Market Information

- Keyplayer Europe

EUROPEAN MARKET SHARE



Others:

- Tachi-S CO.LTD.
- Gentherm Inc.
- Recaro
- Sparco
- Britax Römer
- Maxi-Cosi
- RENNTech Seats
- Groclin SA
- Kongsberg Automotive
- Johnson Controls
- Hyundai Transys

The recliner industry is dominated by a few large suppliers. Adient holds around 30% of the global market share, and the top 10 manufacturers account for roughly 85% of total revenue.

Adient

- 2024 financial report

Item 1. Business

Adient plc ("Adient") is a global leader in the automotive seating supply industry with leading market positions in the Americas, Europe and Asia and maintains longstanding relationships with the largest global automotive original equipment manufacturers ("OEMs"). Adient's proprietary technologies extend into virtually every area of automotive seating solutions, including complete seating systems, frames, mechanisms, foam, head restraints, armrests and trim covers. Adient is a global seat supplier with the capability to design, develop, engineer, manufacture, and deliver complete seat systems and components in every major automotive producing region in the world.

Adient designs, manufactures and markets a full range of seating systems and components for passenger cars, commercial vehicles and light trucks, including vans, pick-up trucks and sport/crossover utility vehicles. Adient operates approximately 200 wholly- and majority-owned manufacturing, assembly or sequencing facilities, with operations in 29 countries. Additionally, Adient has partially-owned affiliates in China, Asia, Europe and North America. Through its global footprint and vertical integration, Adient leverages its capabilities to drive growth in the automotive seating industry.

Adient's business model is focused on developing and maintaining long-term customer relationships, which allows Adient to successfully grow with leading global OEMs. Adient and its engineers work closely with customers as vehicle platforms are developed, which results in close ties with key decision makers at OEM customers.

1. *Core seat structures*: By developing common front seat frames and mechanisms across multiple vehicle platforms, OEMs are reducing costs.

2. *Component sourcing*: Several OEMs have shifted from sourcing a complete seating system to a components approach where the OEM sources each of the different components of the seat and seating assembly as separate business awards.

3. *Engineering "in-sourcing"*: Some OEMs are conducting the design and engineering internally and are selecting suppliers that have the capability to manufacture products on a worldwide basis and adapt to regional variations.

Lear

▪ 2023 financial report

- The company leverages extensive design, material, and integration capabilities as a competitive advantage, supported by the Center for Craftsmanship in Michigan.
- With INTU™ Intelligent Seating, ergonomic, thermal, and adaptive comfort functions are developed to automatically adjust seat parameters and personalize the in-vehicle experience.
- The ConfigurE+™ architecture enables flexible, electrically powered seat positioning without traditional wiring harnesses, ideal for private, autonomous, and commercial vehicle applications.
- A broad technology portfolio includes leather/fabrics, seat mechanisms, zero-gravity seats, and sustainable materials such as ReNewKnit™ and SoyFoam™
- Advanced testing and validation centers ensure high quality, safety, and acoustic performance; the company focuses on sustainability, comfort, safety, and cost efficiency.

The top five customers of our Seating segment are: General Motors, Mercedes-Benz, Stellantis, Volkswagen and Ford.

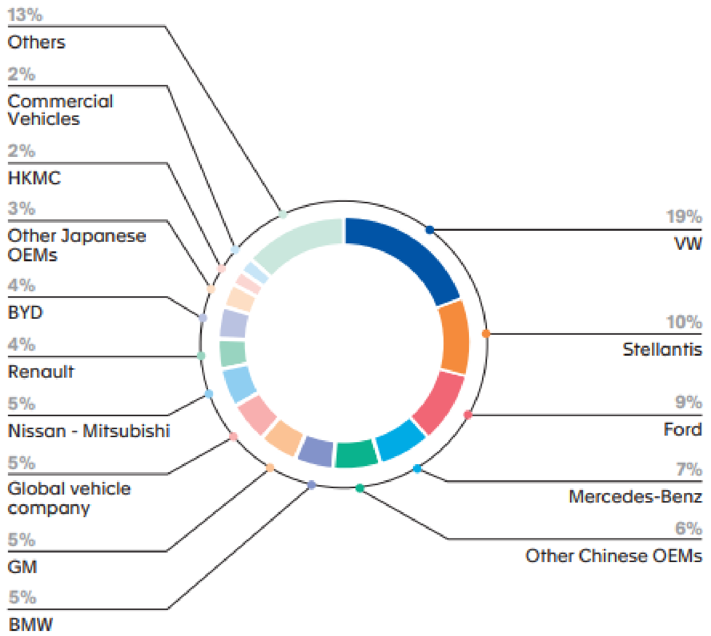
Forvia

- 2024 financial report

(in € million)	FY 2024	Scope effect (1)	FY 2023	Reported	At constant scope & currencies
Seating	8,634.3		8,551.1	1.0%	1.8%
Interiors	5,108.4		4,922.7	3.8%	5.0%
Clean Mobility	4,153.4	(352.7)	4,832.2	-14.0%	-5.3%
Electronics	4,188.5		4,137.9	1.2%	2.6%
Lighting	3,878.7	271.0	3,745.9	3.5%	-2.7%
Lifecycle Solutions	1,010.9		1,058.1	-4.5%	-3.8%
TOTAL	**26,974.2**	**(81.7)**	**27,247.9**	**-1.0%**	**0.4%**

(1) Scope effect includes disposal of CVI, the 100% consolidation of BHAP and the disposal of HUG Engineering.

Sales by customer



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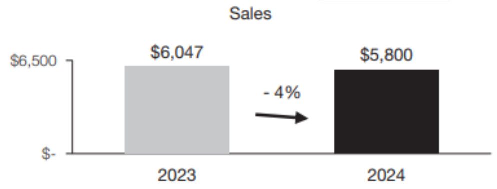
Source; Forvia Financial Report

Magna

- 2024 financial report

SEATING SYSTEMS

	2024	2023	Change	
Sales	**$ 5,800**	$ 6,047	$ (247)	−4%
Adjusted EBIT	**$ 223**	$ 218	$ 5	+2%
Adjusted EBIT as a percentage of sales	**3.8%**	3.6%		+0.2%



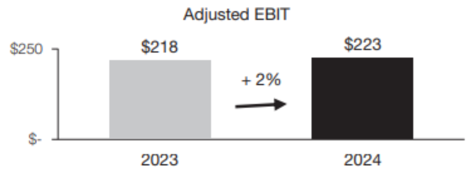
Sales – Seating Systems

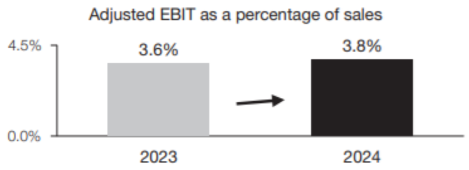
Sales decreased 4% or $247 million to $5.80 billion for 2024 compared to $6.05 billion for 2023, primarily due to:

- the end of production of certain programs, including the:
  - Chevrolet Bolt EV;
  - Ford Edge;
  - Skoda Superb; and
  - Lincoln Nautilus;
- lower production on certain programs, including the Jeep Grand Cherokee;
- the net weakening of foreign currencies against the U.S. dollar, which decreased reported U.S. dollar sales by $18 million; and
- net customer price concessions.

These factors were partially offset by:

- the launch of programs during or subsequent to 2023, including the:
  - BYD Qin L;
  - Mini Countryman;
  - Skoda Kodiaq; and
  - BYD Seal U DM-i;
- customer price increases to partially recover certain higher production input costs; and
- the negative impact of the UAW labour strikes, which negatively impacted 2023 sales by approximately $45 million.





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Source; Magna Financial Report

Summary

Financial reports

Adient remains one of the leading suppliers in the global seating market, while OEMs are increasingly shifting from full seating systems to component-based sourcing.
Forvia recorded slight growth in its seating division in 2024 and maintains strong relationships with key customers such as Volkswagen (20%) and Stellantis (10%), whereas its exposure to U.S./Asian OEMs and commercial vehicles remains limited.
Magna faced declining volumes due to the phase-out of several vehicle platforms but was able to increase its seating-segment profit through price adjustments and production optimizations.

Across the industry, all major Tier-1 suppliers emphasize in their financial reports that they continue to invest in seating systems, both technologically and in engineering capabilities. This strategic focus aligns closely with the **EQYO co-engineering approach**, which relies on early, collaborative system development between OEMs, Tier-1 suppliers, and technology partners.

Market Information

- future outlook

11.12.2025			calculated	given						
recliner markets			calculated							
	car production ww									
	2025	2026	2027	2028	2029	2030	2031	2032	assumptions	
all cars	91.858.383	91.246.596	92.723.463	93.267.956	95.348.316	96.431.610	97.499.395	98.697.222		from IHS data 11-2025
front seats world	183.716.766	182.493.192	185.446.926	186.535.912	190.696.632	192.863.220	194.998.790	197.394.444	2 seats / car, r	from IHS data 11-2025
North and South America cars	18.313.908	18.111.688	19.021.916	19.307.426	20.044.636	20.404.448	20.460.388	20.526.226		from IHS data 11-2025
Asia all cars = Greater China + Japan/Korea + South Asia	54.373.281	53.973.245	54.099.863	54.029.139	54.908.443	55.795.705	56.759.945	57.672.277		from IHS data 11-2025
Europe cars	16.888.562	16.857.108	17.087.162	17.371.639	17.733.307	17.595.108	17.712.695	17.937.283		from IHS data 11-2025
middle East / Africa cars	2.282.632	2.304.555	2.514.522	2.559.752	2.661.930	2.636.349	2.566.367	2.561.436		from IHS data 11-2025
number of sliding elements/car N and S America	73.255.632	72.446.752	76.087.664	77.229.704	80.178.544	81.617.792	81.841.552	82.104.904	4	2 recliner/seat*1 bearing/recliner*2 seats = 4bearings/car
number of sliding elements/car Asia all	217.493.124	215.892.980	216.399.452	216.116.556	219.633.772	223.182.820	227.039.780	230.689.108	4	
number of sliding elements/car Europe	42.221.405	42.142.770	42.717.905	43.429.098	44.333.268	43.987.770	44.281.738	44.843.208	2,5	sometimes 2 (Brose) and rest has 1 bearing/recliner
number of sliding elements/car in Middle East / Africa	4.565.264	4.609.110	5.029.044	5.119.504	5.323.860	5.272.698	5.132.734	5.122.872	2	lower number assumed, minor effect
all sliding elements ww without 2nd and 3rd seat row	337.535.425	335.091.612	340.234.065	341.894.862	349.469.444	354.061.080	358.295.804	362.760.092		
turnover potential ww all	101.260.628	100.527.484	102.070.220	102.568.458	104.840.833	106.218.324	107.488.741	108.828.027	0,3	€/pc assumption, only
EQYO market share all won ww on volumes	< 3%									
EQYO market share all HP won ww on volumes	< 1%									

The global recliner market is set to grow steadily in line with automotive production. While penetration is already near 100%, market value increases are driven by technology upgrades, especially the rising share of electric recliners. Revenues are forecast to expand from about $3.5 billion in 2024 to roughly $5.2 billion by 2033 (CAGR ~4–5%), with unit volumes growing more moderately (~2–3%). Demand remains stable through 2030, supported by growth in Asia and replacement needs in mature markets

Market Trend

- OEM Feedback

The share of electric recliners is increasing currently ~40% compared to 60% manual recliner, driven by premium vehicles and e-mobility.
Electric vehicles and autonomous driving raise system requirements for example higher weight and integrated seatbelt systems which lead to lighter and more robust recliner designs.
Loads are increasing while cycle requirements remain unchanged.

Excerpt from the customer visits Q1/2026:

BMW unfortunately does not see a rising load or wear specification for their seats/recliners.
For that reason, they did not show further interest in HP for recliners.

Audi

The increasing number of functions in the backrest leads to higher overall weight.
→ This will increase the load requirements for the recliner.
→ Lifetime and cycle requirements are expected to remain unchanged.
The recliner width is intended to remain unchanged.
A smaller radius would theoretically be possible but is currently not part of any active development.

Market Trend

▪ Internal interviews - Key Insights

- Historically a major producer (40 million RC70 units); restarting production in China.
- PRO100E is too expensive → CSB currently supplying DU (0.10–0.20 €/pc).
- New RC90 generation (since 2020 in China) → higher lifetime and EV requirements.
- Two potential bearing positions (Ø28.5 polymer outside / Ø52 polymer inside).
- Target price PFAS-free ≤ 0.35 €/pc.
- Currently no business in recliner applications.

FORVIA

Market Trend

▪ Internal interviews - Key Insights Europe

- RFQs for 1–10 million pcs/year, realistically 7–8 million pcs/year.
- Our pricing ~3× higher than competitors (GGB/CSB).
- Design width increased: 5 mm → 7 mm → 10 mm.
- New load requirements: 2250–3000 N (previously 1500 N), dynamic 3–4 g.
- Competitor likely GGB (17–20 ct/pc).
- PRO100E performed 20% better in testing, but price prevented supplier change.
- No current business in Europe.

Market Trend

- Internal interviews - Key Insights Europe

- Brose Inquiry volume: 140 million pcs / 8 years ≙ 17.5 million pcs/year.
- Require 6 geometries, polymer inside/outside, often only 1.25–1.5 mm thickness → declined by EQYO.
- Lifetime testing: PRO100E & HPPRO100GG achieve only 10% of the requirement.
- Competitor KS with P141, P209, P243 (cross grooves + Surtec533 grease) reaches 50–60%.
- Brose currently accepts performance deficits due to lack of alternatives.
- Future requirement: >60% lifetime, PFAS-free.

brose

Market Trend

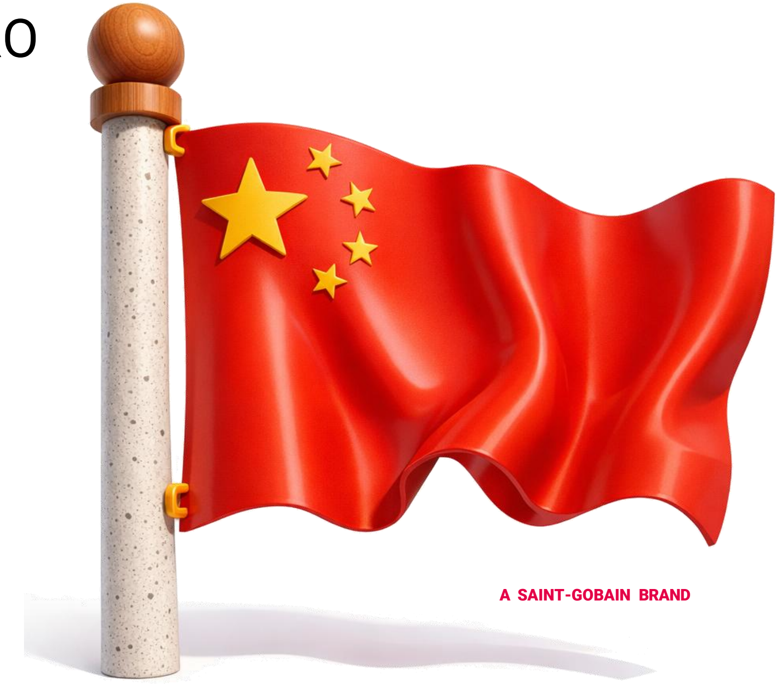
▪ Internal interviews - NA Market (Lear, Fisher, Camaco)

- Market size: >60 million recliners/year (2 per seat).
- ID range: 36–40 mm.
- Successes: 7 million PRO100E + 0.5 million HPPRO100GG at Fisher.
- Lifetime increased to 8000 cycles (previously 3200 like Japan → PRO suitable there).
- Lear: stiffer seat frame → varying stiction left/right; 4 kN test passed.
- Pricing issue: HP too expensive — desired: material between PRO & HP (better friction + better wear at PRO price level).
- Fisher prefers high performance → higher potential for HP.
- Competitors: DU (GGB) 21–28 $ct, we ~40 $ct.

Market Trend

▪ Internal interviews - China (Keiper/Yanfeng, Lear, Adient, Lile)

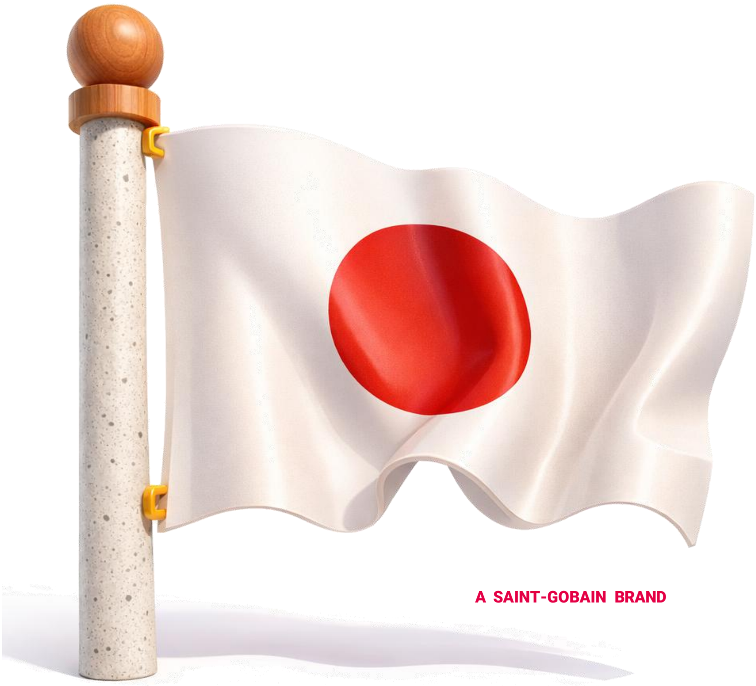
- Market: 24 million vehicles → ~96 million recliners/year.
- ID mostly ≤ 20 mm, one bearing per recliner.
- Competition dominated by GGB (lowest price), CSB, Mingyang.
- Typical pricing: 0.10–0.15 €/pc, we are 3× higher (HP).
- Minimal business (SAIC 120k/year, some Geely).
- BYD → no success despite in-house recliner production.
- Opportunities: 4 mm instead of 6 mm width, higher pv capability (PRO ~2.5; HP 3–4 MPa·m/s) → relevant for lightweighting & downsizing.



Market Trend

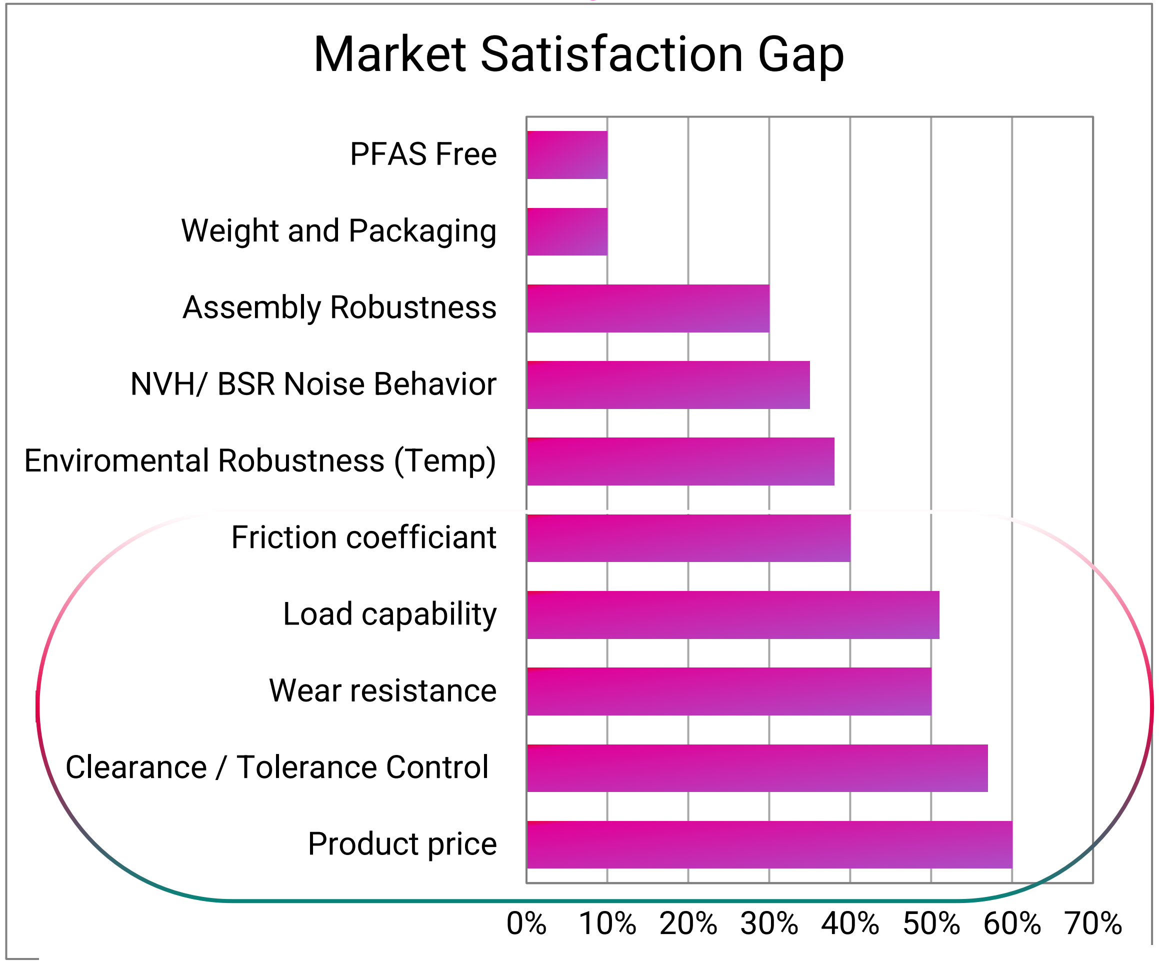
▪ Internal interviews - Japan (Toyota Boshoku / Imasen)

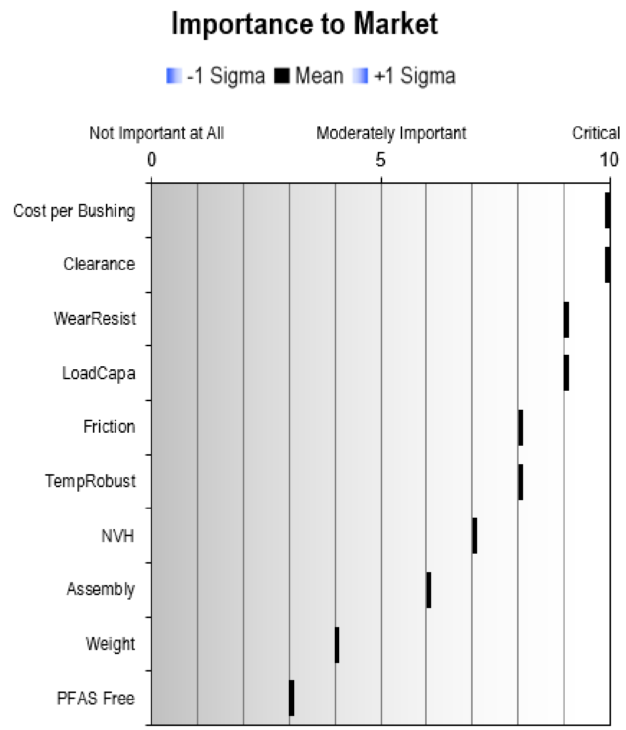
- Currently using PRO050E in 7 models; failures observed in China (likely welding issue).
- HPPRO050GG (single-flange) proposed to generally improve performance.
- Loads increasing due to: zero-gravity seats, heavier vehicles, lightweight construction → higher surface pressures.
- Imasen needs downsizing due to lightweighting — opportunity for HP materials.
- If DU works → almost no chance to win business in Japan.

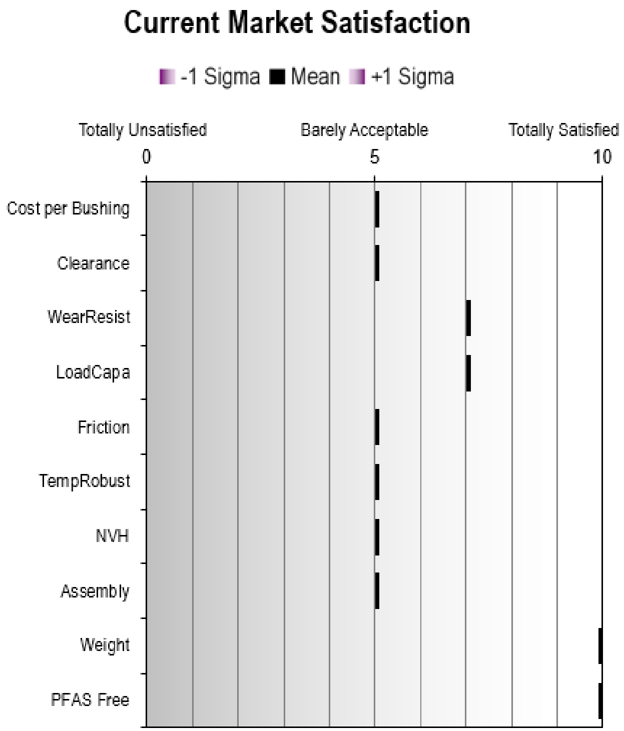


Voice of Customer

Technical Challenges and Market Satisfaction Gap

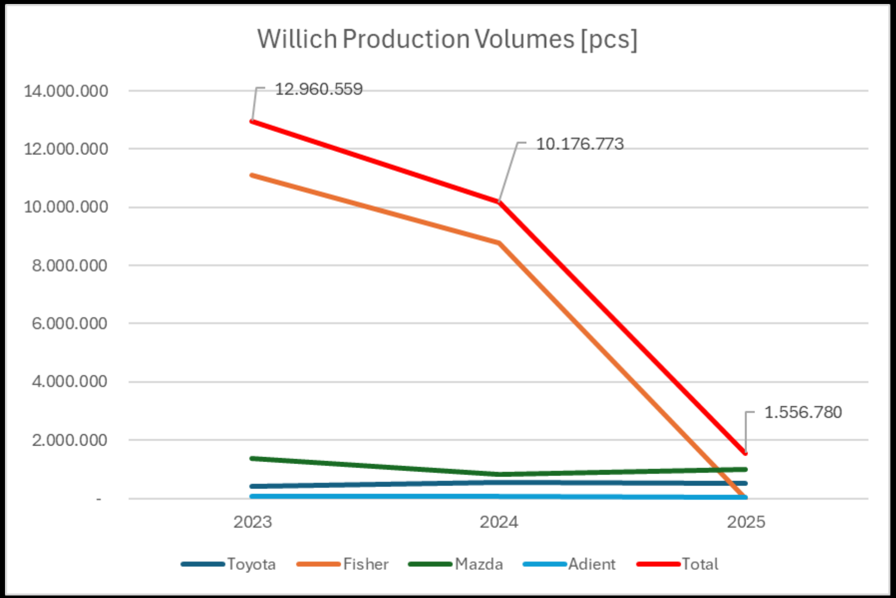


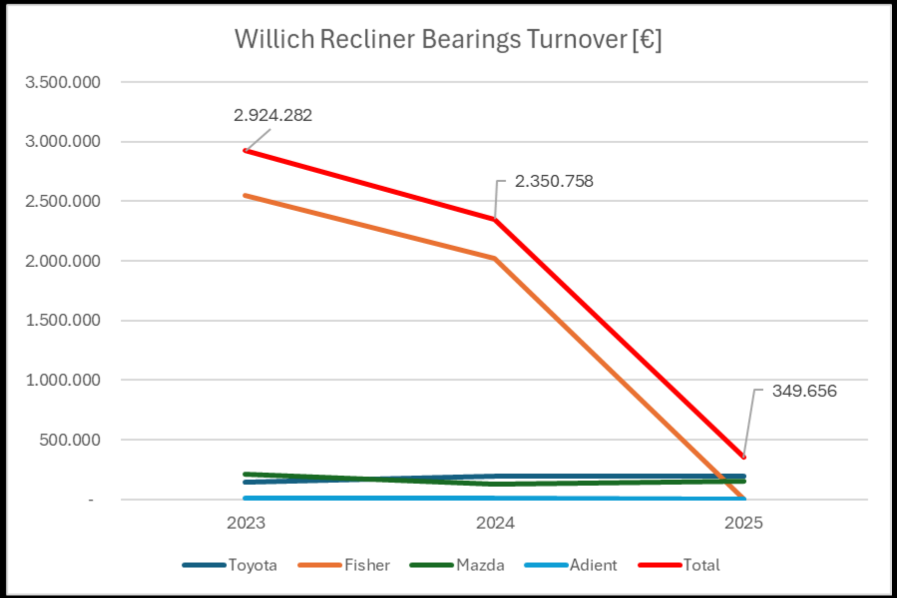




Market Information

Willich business





Value Chain + Key Players

- Safety mechanisms
- Airbag integration
- Innovative safety solutions

Autoliv

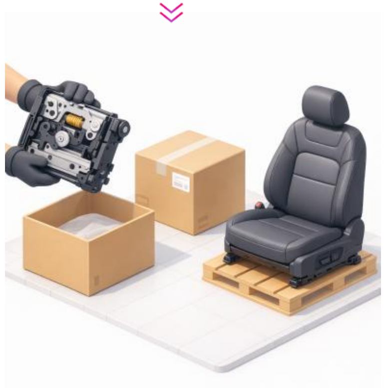
- Front and rear seat frames
- Metal structures
- Seat mechanisms (tracks, recliners, adjusters)

ADIENT
Diniz Adient Oto Donanım

- Foam
- Fabrics
- Metal components

ADIENT

- Seat architecture & design
- Ergonomics development
- Crash behavior & safety
- Comfort functions (heating, ventilation, massage)
- Electronics and sensor integration
- Lightweight construction and sustainability concepts



- Front seats
- Rear seat benches
- Seat modules (e.g., backrest, seat cushion, mechanisms)
- JIT deliveries to the vehicle assembly line,
- Provide individual components to other TIER1/ OEMS

Benchmarking and Side by Side Testing

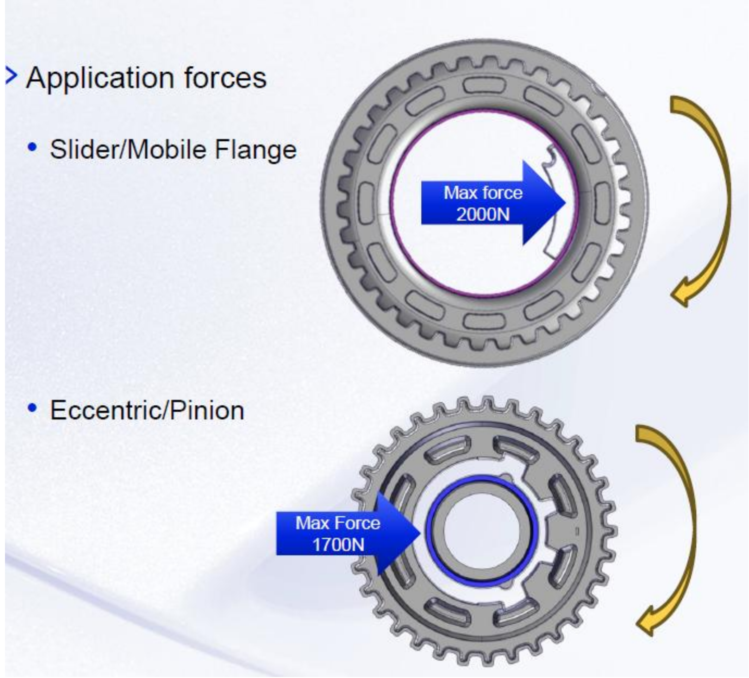
▪ Forvia RC90 Recliner

- Temperature
- Operating range: -45°C to +93°C
- Seat max. storage temp: +120°C
- Exceptional temperatures :
- >200°C during welding (max 30s)
- +210°C for 1h during cataphoresis application

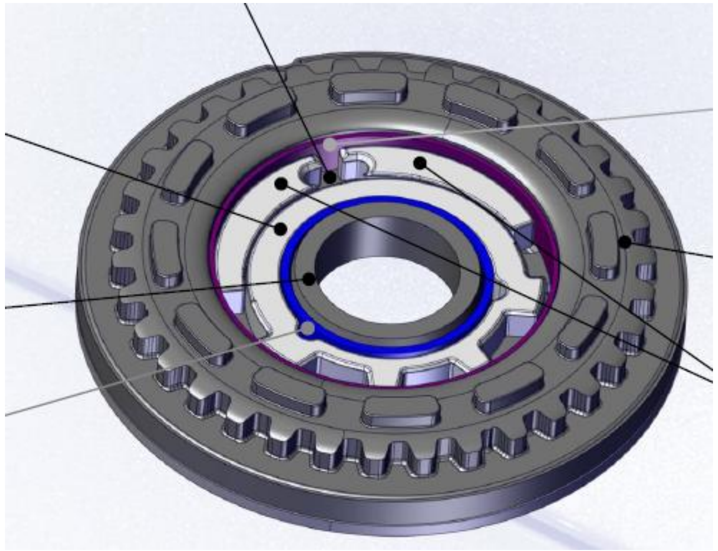
- Grease
- Runs with grease

- Operating speed
- From 20 to 60 rpm (command)

- Durability
- 1 cycle = 30 turns
- 4000 cycles @RT
- 1000 cycles @-20°C
- 1000 cycles @+60°C



Large bearing
Ø52x height 5mm



Small bearing
Ø28,5x height 5,5mm

Benchmarking and Side by Side Testing

- Forvia RC90 Recliner

Competition:

- 001
- 002
- 003
- 006
- 007
- 008

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Benchmarking and Side by Side Testing

- Forvia RC90 Recliner

Benchmarking and Side by Side Testing

- Brose

- Friction
- Bushing: ø 38xø41x6mm
- Velocity: 105°/sec
- 1 cycle: ±360°
- Load: 1500N
- Temperature: RT

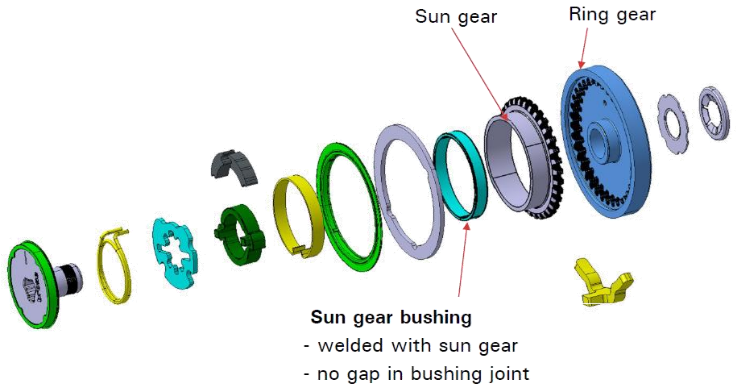
Friction over time

- Grease
- Grease lubrication: provided by Brose

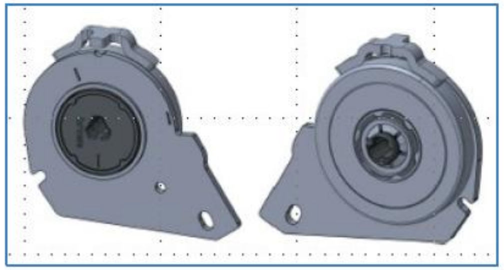
- Operating speed
- Frequency: 0,1 Hz (144°/s)

- Durability
- 1 cycle: ±720°
- Load: 3000N
- Duration: 20h and 48h
- Temperature: RT

Torque over time

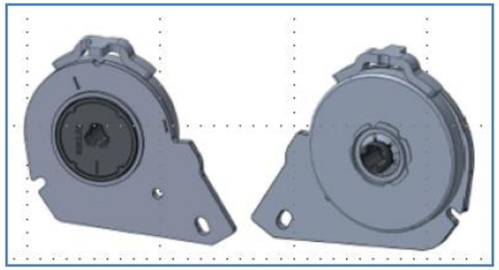


LDB 2000 (1st generation)



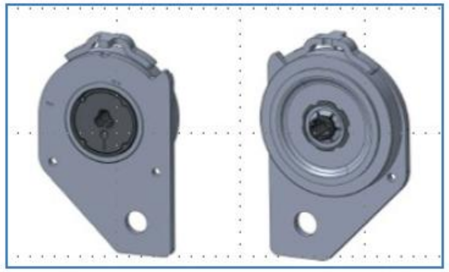
LDB 2000
- Load class: 2,000 Nm
- construction space (ØxH): 70 x 11.1
- power & manual

LDB2700 (2nd generation)

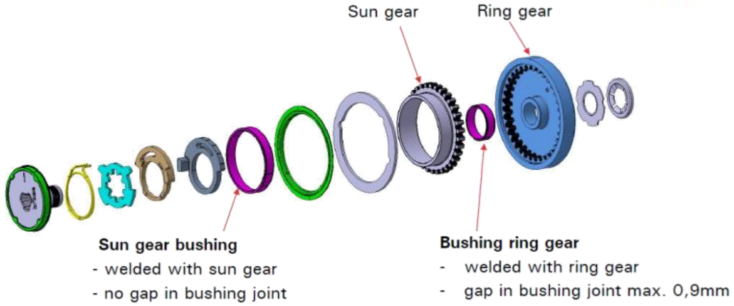


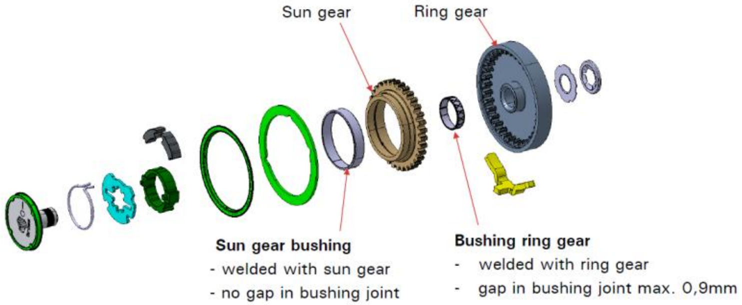
LDB 2700
- Load class: 2,700 Nm
- construction space (ØxH): 70 x 10
- power

LDB 6300 (rear seat)



Rear seat
- Load class : 6.300 Nm
- construction space (ØxH): 81 x 13.6
- power





Benchmarking and Side by Side Testing

- Adient Taumel 3000 Recliner

Recliner – Bearing development

ADIENT

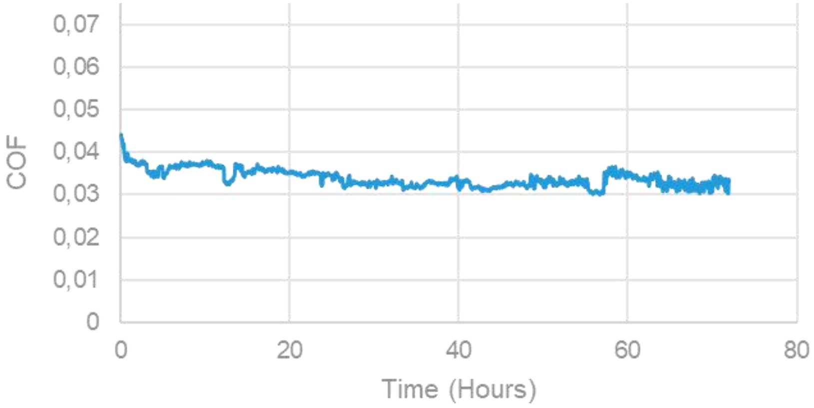
Validation plan for bearing:

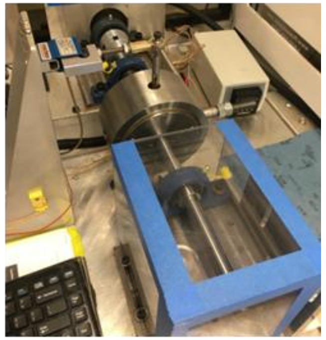
1) The load on the bearing is it be on a projected surface as applied by the wedge. The wedge / Omega spring and Cam will be provide.
2) Request test with applied load of 1600N over bearing surface with the wedge. Also request to understand the bearing performance with 0.5X , 1X and 2X (base 1600N) load through the wedge system.
3) Existing recliner operation speed is ~120°/s at bearing and the expectation is the speed would increase between 4 to 8 times. Request the understand which bearing would suit such speeds and request test results to understand bearing wear at 1X, 4X and 8X test speed (120°/s – 1X)
4) Number of cycle the bearing can fulfil at different load and speed condition.
5) Performance at extreme temperature (-30° and +80°C)
6) Test condition (with / without lubricant) – If with which lubricant used. Current grease will be provided for testing.

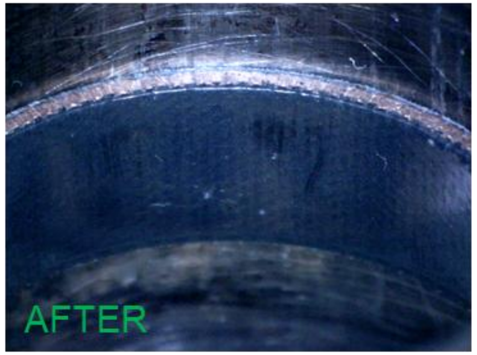
Benchmarking and Side by Side Testing

- Adient Taumel 3000 Recliner

Conditions
Material: HP100GG
Grease: FUCHS
Pressure: 22.4MPa
Surface Speed: 0.13m/s
Duration: 72h



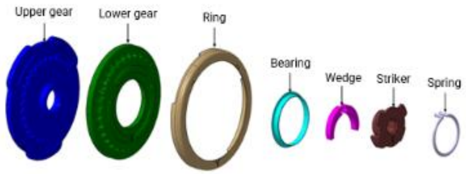
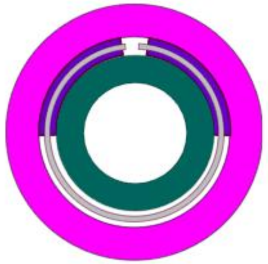
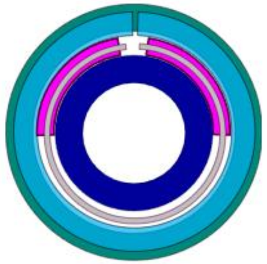


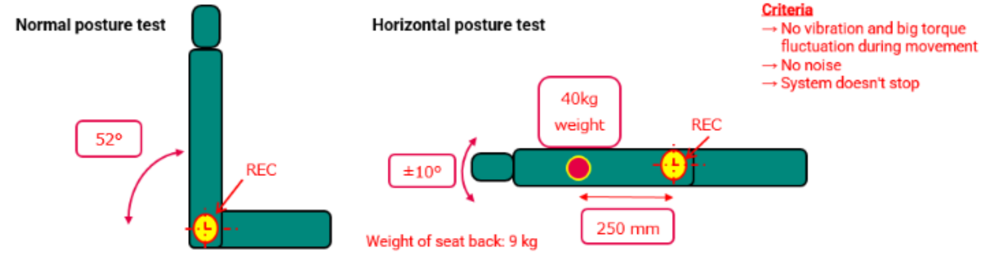


Average COF = 0.0338
Total Wear ~ 9µm

Benchmarking and Side by Side Testing - Japan

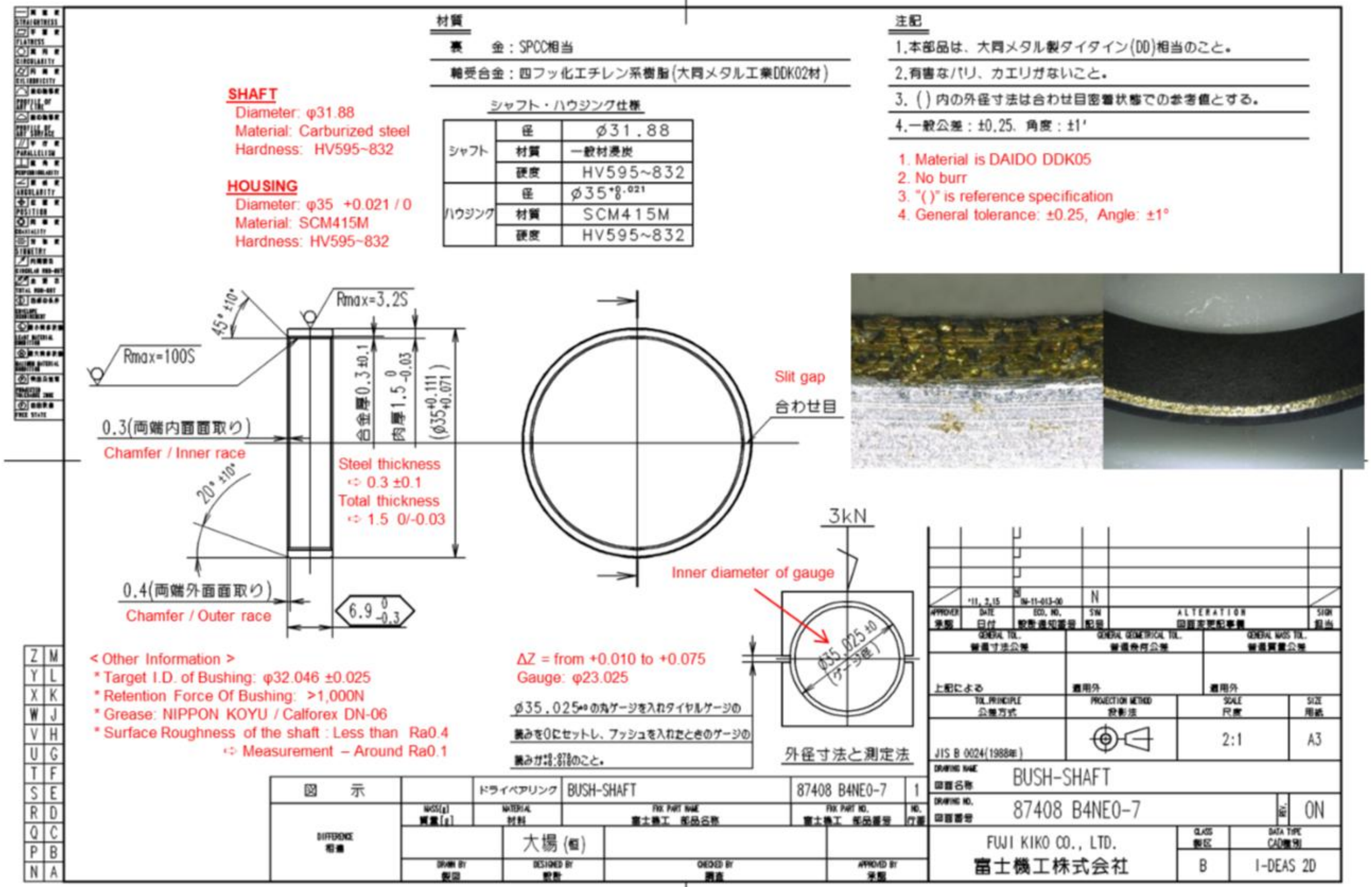
- Toyota Boshuko

Type of recliner	Standard seat recliner	High strength seat recliner
	Mainly used for front seat	Used for seat belt build-in type seat. Mainly for rear seat
Feature	No bush design. The wedges are mirror finishing by barrel process. Welding points are closer than high strength seat recliner that have welding heat issue on PRO100E.	They used to use two standard seat recliner for one seat for the seat that needs high strength before they developed high strength seat recliner. Currently. One high strength seat recliner operate reclining
Image		
Structure		
	 Wedge · Mirror finishing of the wedges by barrel process	 Wedge · Radius: R18 · Thickness: 5.8 mm Bush · PRO050E now · Will switch to HPPR O050GG to improve heat resistance.
Annual volume	3.4M pcs	510K pcs

Test condition / Assembly		
Durability test	Normal posture → Number of cycle: 13500 cycles → Angle: 52° Horizontal posture → Number of cycle: 1500 cycles → Angle: ± 10°	Normal posture → Number of cycle: 6750 cycles → Angle: 52° Horizontal posture → Number of cycle: 750 cycles → Angle: ± 10°
		
High / Low temperature test	Temperature: -30 degreeC and +80 degreeC Number of cycle: 500 cycles Criteria +80 degreeC condition → No torque difference before and after test. → No abnormal movement or NVH -30 degreeC condition → No functional issues	
Estimated load, speed and sliding distance of wedge	Estimated surface load: 130 MPa Speed: 40 mm/sec Sliding distance of wedge: 6000 m	Estimated surface load: 110 MPa Speed: 35 mm/sec Sliding distance of wedge: 3500 m
		Reference: Measured surface load to the bearing · Without load → Surface load: 35 MPa (by pressure-sensitive paper) → Speed: 37.3 mm/sec · With 40kgf load → Surface load: 102 MPa (by pressure-sensitive paper) → Speed: 32 mm/sec

Benchmarking and Side by Side Testing - Japan

- Imasen



Benchmarking and Side by Side Testing - Japan

- TF Metal

TF METAL (former FUJI KIKO)
https://www.tf-metal.jp/en/



Main customer: NISSAN
Volume: 1,920K pcs (Japan:80Kpcs/m, China:80Kpcs/m)
Current solution: Daido Metal DDK05 → 20 JPY/pc (0.1163 EUR)
Drease: NIPPECO Calforex DN-067

Durability cycle
・1 cycle = CW 9 rotations → CCW 9 rotations
・Durability: 8000 cycles
・Surface load to the bearing during the test: 86MPa
・Speed: Speed: 0.00209m/s (8 sec = 60°)

Others
・Retention force: over 1000N after assembly

Information
・Wear issue on DU bearing

Scorecard

Opportunity	Opportunity Size						Comments
	<500K	<1M	<5M	<10M	10-20M	20M+	
TAM / SCALE POTENTIAL						X	
	1 Cust	1-2 cust 1 region	Most cust 1 region	1-2 global cust	Most global cust / 2-3 regions	All global cust/ . reg	
OPPORTUNITY SCALE						X	

Risk Reduction	Opportunity	Evidence and Confidence						What experiment can we do to reduce risk?
		No Evidence	Light Evidence (Read)	Light Evidence (say)	Evidence from multiple sources (say)	Light call to action (do)	Strong Call to action (do)	
VALUE PROPOSITION	We have identified a true pain in a critical customer segment		X			X		
CUSTOMER DISSATISFACTON	The customer is dissatisfied enough to be motivated to overcome the pain					X		
CUSTOMER RELATIONSHIP	We have the right relationships with the right customers to bring innovation to the market						X	

Scorecard

Risk Reduction

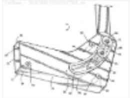
		Evidence and Confidence						What experiment can we do to reduce risk?
		No Evidence	*Light Evidence (Read)*	*Light Evidence (say)*	*Evidence from multiple sources (say)*	*Light call to action (do)*	*Strong Call to action (do)*	
	Feasibility							
KEY TECHNOLOGY	We have the right technologies to develop a solution to the value proposition		X					
KEY CAPABILITIES	We have the right people, resources or partners to develop our value proposition						X	
	Markets and Competition							
MARKET TRENDS	The customer's pain points are within technologies positioned to benefit from key trends						X	
COMPETITION	Our benchmarking shows that we have a differentiation over competition + we can protect this.				X			
	Viability							
TCO / PRICE	The customer is willing to pay the price of the value proposition, considering TCO	X						
COSTS	The costs of creating the value proposition allows to create sufficient contribution margin				X			

Patents

- Research

Showing records 1-10 of 399 | Sort (Relevance) | Analyse | Optimise search | Export | To folder

1) US2005017561 AA - SEAT, SEAT RECLINER MECHANISM, AND SEAT RECLINER SYSTEM



Abstract: The seat back of a seat is attached to a seat recliner mechanism. As the seat back is moved, it moves in a curvilinear motion centered about or near the seat H-Point. The seat recliner mechanism, attached to the seat back, has arcuate slots. The center of the curves of these arcuate slots are located at or near the seat H-Point. A seat recliner system has two seat recliner mechanisms operating in tandem.

AI Summary

Probable assignee: EXCELLENCE MFG INC

First published: January 27, 2005

2) US2002175548 AA - SEAT RECLINER

Abstract: The seat recliner includes a first seat member. The seat recliner includes a second seat member rotatable relative to the first seat member. The seat recliner includes an input cam member for rotating relative to the first seat member. The seat recliner includes a pivot on the first seat member. The seat recliner includes a follower member for being actuated by the input cam member to lock the second seat member relative to the first seat member. The follower member includes a first side. The first... (Read more)

AI Summary

Probable assignee: TF METAL CO LTD

First published: November 24, 2002

3) US2007278836 AA - SEAT RECLINER FOR VEHICLES

Abstract: Disclosed herein is a seat recliner for vehicles. The seat recliner of the present invention includes a support bracket, which is mounted to a seat frame, a guide holder, which is coupled at a first end thereof to the support bracket, and a slide holder, which is rotatably coupled to the guide holder between the support bracket and the guide holder. The seat recliner further includes at least one rotating bracket, which is coupled to the slide holder, a clutch unit, which controls the engagement between the guide holder and... (Read more)

AI Summary

Probable assignee: AUSTEM CO LTD

First published: December 5, 2007

Optimise search

box(es) twice. If an entry is not selected, they might still appear in the final search set.

Companies

Select companies

- Fisher Dynamics Corp. (26)
- Hyundai Transys Inc. (17)
- Lear Corp. (17)
- Hyundai Motor Co. Ltd. (10)
- Fisher And Co. Inc. (9)
- Jiangsu Lile Automobile Components Co. Ltd. (8)
- DAS Co. Ltd. (6)
- Austem Co. Ltd. (5)
- BAE IND Inc. (4)
- Daewon Prec IND Co. Ltd. (4)

Optimise Search

- Technologies - IPC
- Technologies - CPC
- Publication dates

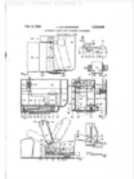
Patents

- Research

Search: TA=(Seat backrest adjustment) | Refine search

Showing records 1-10 of 54 | Sort (Relevance) | Analyse | Optimise search | Export | Save

1) US3020088 A - AUTOMOBILE FRONT SEAT BACKREST ADJUSTMENT



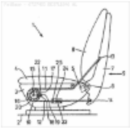
Abstract: (Claim 1) rest 54, here shown in a normal generally vertical position, the lip 46 on the end of arm 45 is moved beyond the end of the are portion of member 47, as indicated by the chain lines in FIG.

AI Summary

Probable assignee: JULES HALTENBERGER

First published: February 6, 1962

2) DE3711846 A1 - Vehicle seat backrest adjustment



Abstract: The vehicle seat (1) has a backrest adjustment arrangement (9) in which the respective free ends (13, 14) of the two tension members (11, 12) are fixed on both sides of the axis of rotation (6) to the backrest (7), whereas the other sides are guided on an operating element (10) which can be locked by a detent pawl (18).

AI Summary

Probable assignee: BAYER AG

First published: October 27, 1988

3) DE19729562 A1 - Vehicle seat backrest adjustment gearing

Abstract: The mechanism to adjust the backrest of a vehicle seat has a layer of sliding varnish on at least one of the cogwheels and/or at least one of the gap surfaces and/or on a wedge surface of the component parts, which also acts as a lubricant. The lubricant has an agent to give a plastic distortion of the coating, such as a dithiophosphate, or a metal dialkyl-dithio-phosphate where the metal is molybdenum, lead and/or zinc. The sliding varnish coating is of poly-tetrafluoroethylene (PTFE), graphite and/or molybdenum sulphide.

AI Summary

Probable assignee: VOLKSWAGEN GROUP

First published: January 22, 1998

Optimise search

It is possible to narrow down a search set to include or exclude selected entries. Include entries by clicking on the relevant box(es) once, exclude entries by clicking on the relevant box(es) twice. If an entry is not selected, they might still appear in the final search set.

Companies

Select companies

- Zhao Cited (7)
- Ciar SPA (2)
- WU YIN (2)
- Alfa Romeo SPA (1)
- Avic Aerospace Life Support IND Ltd. (1)
- Bayer AG (1)
- Beijing Automobile RES General Inst Co. Ltd. (1)
- Bertrand Faure Sitztechnik GmbH And Co. KG (1)
- BYD Co. Ltd. (1)
- Changchun Faway Adient Automotive Syste... (1)

Optimise Search

Conclusion

From the market case, it becomes clear that the demand for modern recliner systems continues to grow. Currently not for PFAS free ones.
OEMs and Tier1 suppliers are particularly focused on materials that provide increased load capacity while offering a stable coefficient of friction in the range of ~0.04 to 0.07 (in lubricated condition). Also increased wear and load resistance are features to be improved (Brose case, NA demands, China). At the same time, the need for thicker bearings is rising, especially in recliner designs with welded connections, where wall thicknesses between 1.25 and 1.5 mm are increasingly becoming the standard.
The market continues to be strongly dominated by DU type products, which represent serious competition both in terms of price and performance. For our own product solutions, this means that technological advancements, material optimizations, and a clearly communicated added value over established alternatives will be essential to gain and secure market share in the long term.
Attractive as well is simply the market size with more than 40 million sliding elements in EU, only. EQYO has < 3% market share (in the current situation – loss of Fisher business < 1%) worldwide. Cost levels for potentially new products must be extremely well monitored.

- Material cost ~≤ Pro
- Thickness up to 1,5mm, (0,5, 1 & 1.25mm as well)
- COF Range ~0,04 -0,07 in lubricated
- Closed Gap*

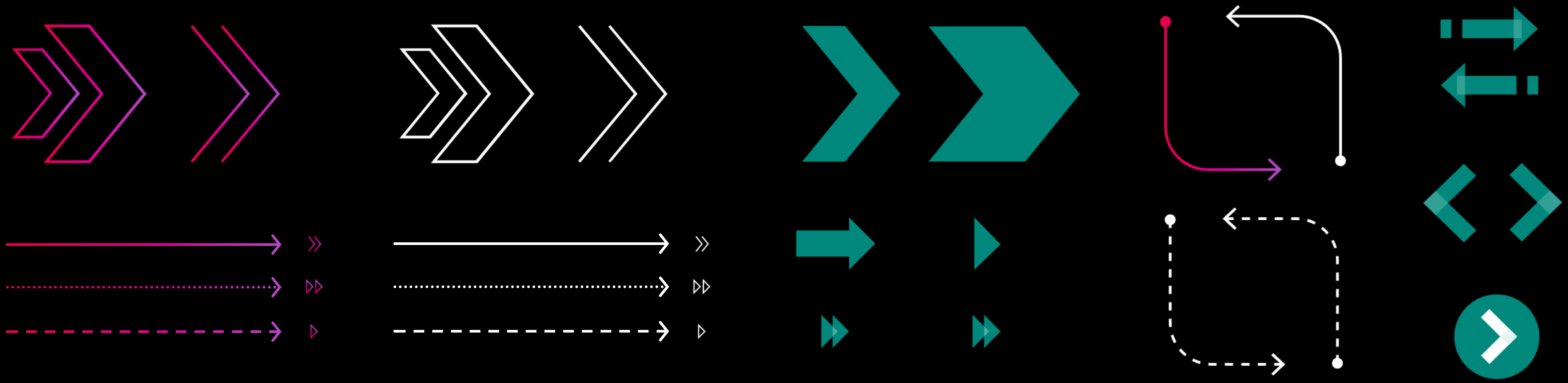
KEY BENEFITS

- Movement
- Protection
- Energy
- Size
- Time
- Process
- Performance
- Speed
- Pressure
- Temperature
- Leakage

KEY BENEFITS – PLEASE SELECT FROM:

ICON Name (internal use only)	ICON Image (copy & paste)	Benefits	ICON Name (internal use only)	ICON Image (copy & paste)	Benefits	ICON Name (internal use only)	ICON Image (copy & paste)	Benefits
Longevity		• HIGH DURABILITY • ROBUSTNESS • FAST & ACCURATE • RESPONSE TIME • LOW WEAR	Temperature		• HIGH TEMPERATURE • CRYOGENIC TEMPERATURES • EXTREME TEMPERATURES	Manufacturing		• EFFICIENT ASSEMBLY
Movement		• CONSISTENT FRICTION & TORQUE • CONSISTENT SLIDING FORCE • ROTARY & LINEAR MOVEMENT • LOW FRICTION • NO STICK-SLIP	Leakage		• LOWEST LEAKAGE • NO PERMEATION • LOW LEAKAGE	Things fitting together		• TIGHT TOLERANCES • SECURE FASTENING • NO FREE-PLAY
Protection		• CORROSION FREE • CHEMICAL RESISTANCE • HARSH CHEMICALS • RESISTANCE	Pressure in the system		• HIGH PRESSURE • ULTRA-HGH VACUUMS	Sustainability	PFAS free	• PTFE-FREE COMPOUND LAYER
Improved system		• MAINTENANCE-FREE • SIMPLIFIED SYSTEM	Energy		• LOW ENERGY CONSUMPTION • ENERGY-SAVING	NVH		• REDUCED NOISE • REDUCED VIBRATION
Speed of our components/ applications		• HIGH SPEED AND ROTATION	Size		• SAVE SPACE • DOWNSIZING			

ARROWS - PLEASE SELECT FROM



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1

2

3

4

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Bus qui ommod quisimint dolorem quatius, nis enias endande bisapis

Description to go here

bisquundit que plab iminven tionsed eum harunt aut mi, ant, im et voluptatt amus sum et ut alicil minverum ium et qua

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